

# Moderation of Hormonal Influences on Women's Extra-Pair Sexual Interests: Two Preregistered Studies

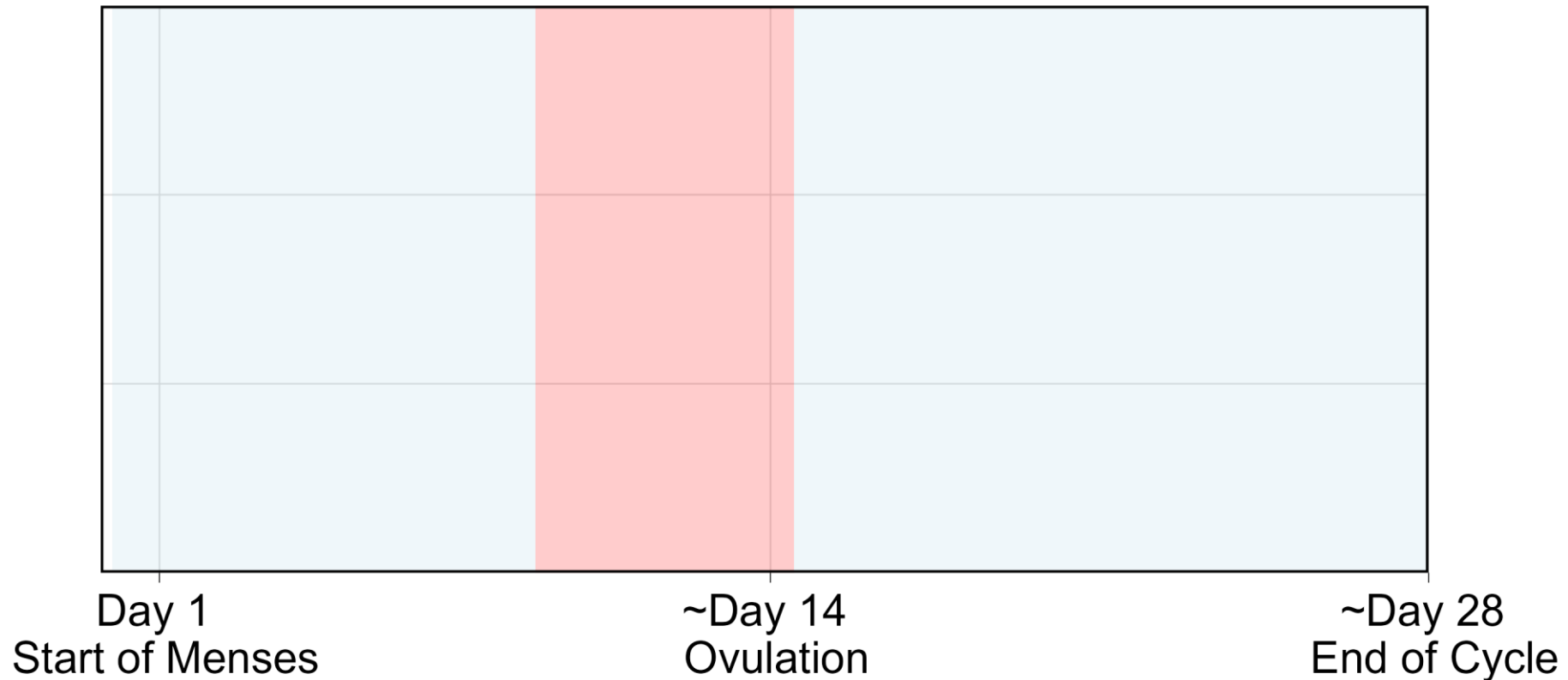
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AREA

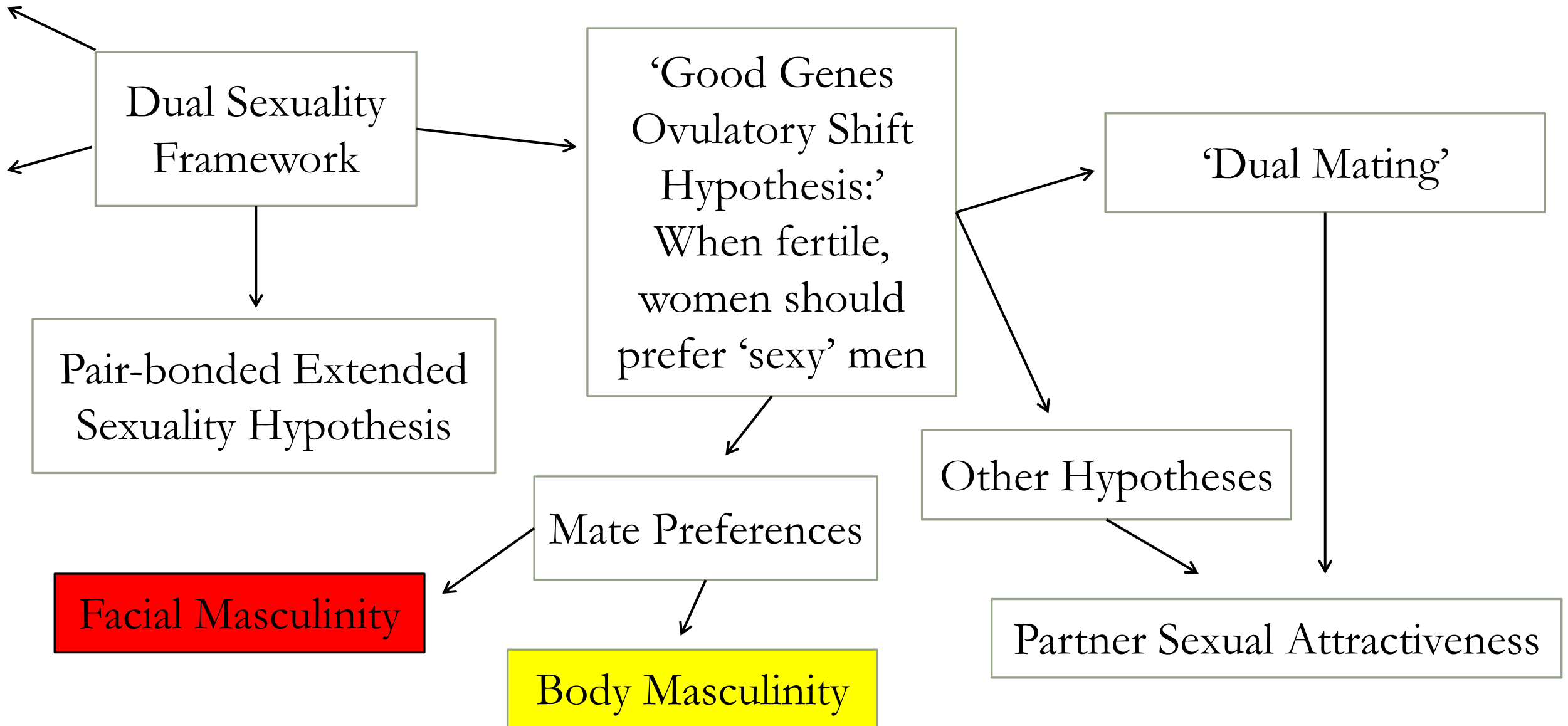
# Dual Sexuality Framework

 Estrous Sexuality  Extended Sexuality



If specific fitness benefits or costs of sex differ across phases of the cycle, then selection should favor differences in the systems that regulate sexuality across phases.

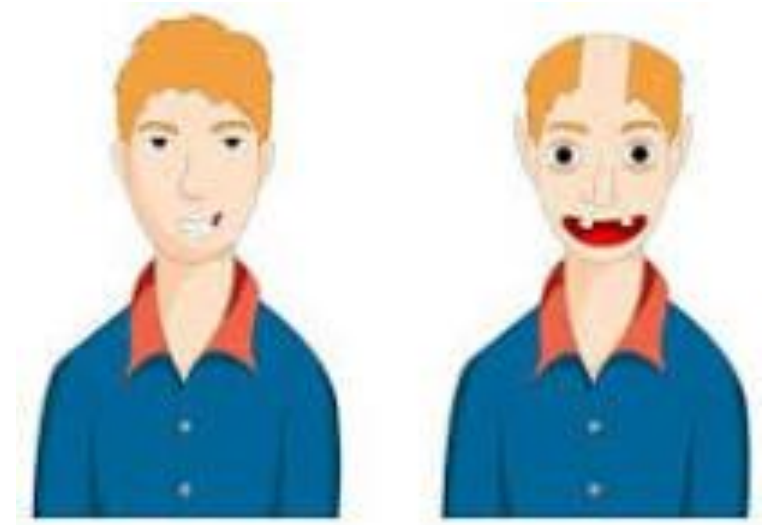
# Deriving Hypotheses from Dual Sexuality Framework



# Partner Sexual Attractiveness Moderation

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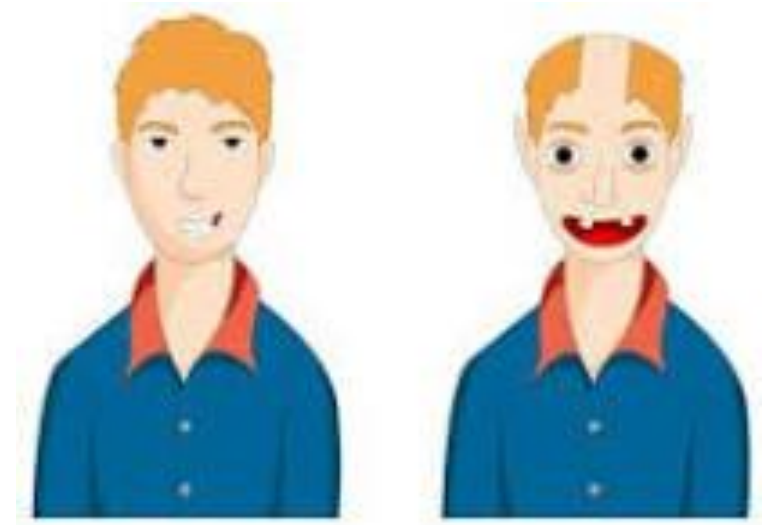
➤ “If women’s preferences for ‘sexy’ features are elevated when conceptive, then women might experience a shift in attraction to extra-pair men—but only if their primary partners are relatively low in sexual attractiveness” (pg. 715, Gangestad et al., 2023).



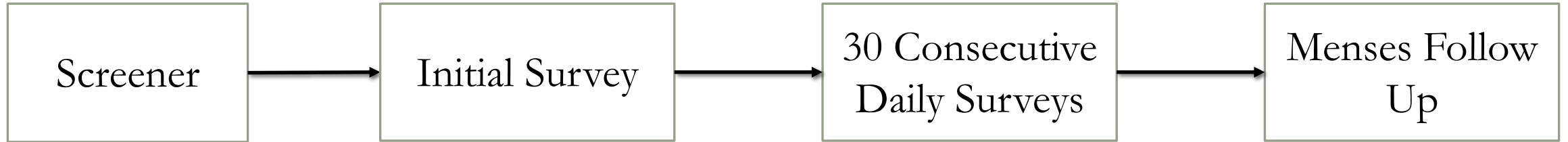
# Partner Sexual Attractiveness Moderation

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- Early support (e.g., Haselton & Gangestad, 2006; Pillsworth & Haselton, 2006).
- Within-woman design found consistent evidence (Arslan et al., 2021; Gangestad & Dinh, 2021).
- Preregistered lab study (Dinh et al., 2022).
- Non-preregistered analysis found negative evidence, but their outcome fails basic validity test (Schleifenbaum et al., 2024).

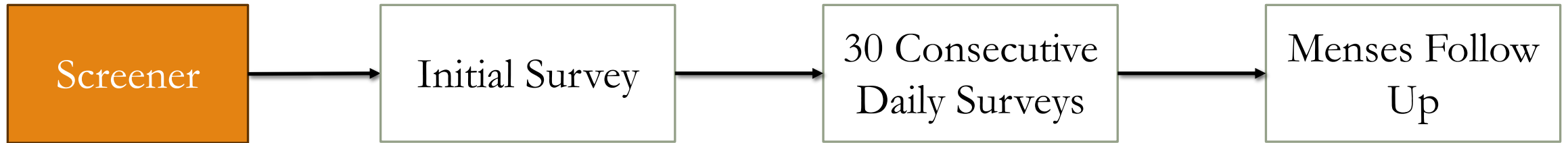


# Study Design



- Two daily diary studies administered via Prolific
  - Study 1: spring 2024 (up to \$91)
  - Study 2: spring 2025 (up to \$53)
- All methods were preregistered for Study 1 and Study 2

# Study Design

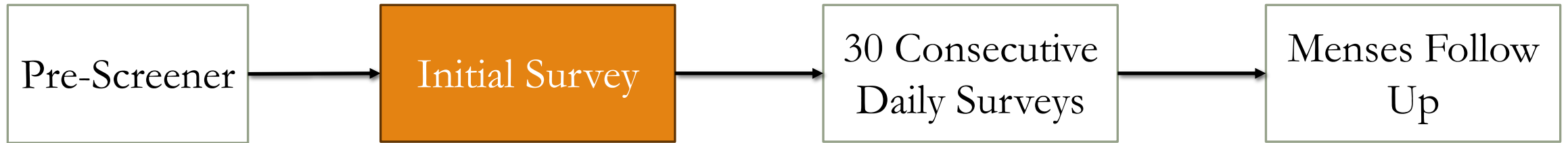


## Inclusion criteria:

- In a heterosexual romantic relationship (at least 3 months long)
- Ages 18-35
- Naturally cycling
- Experienced regular cycles (not varying more than 5 days in length across months)
- Living in U.S. or U.K.

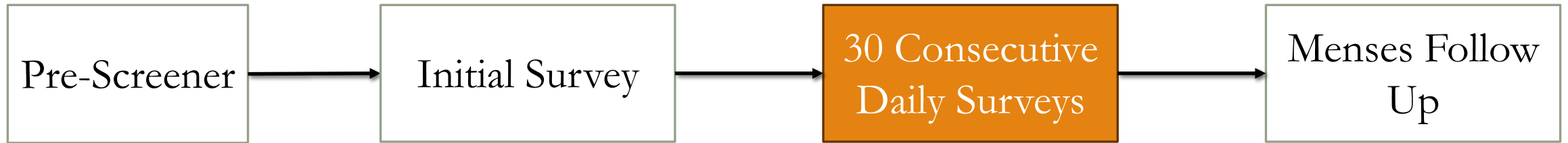
317/307 (Study 1/2)  
met initial criteria and  
were invited to  
participate in a study on  
“Relationship  
Dynamics”

# Study Design



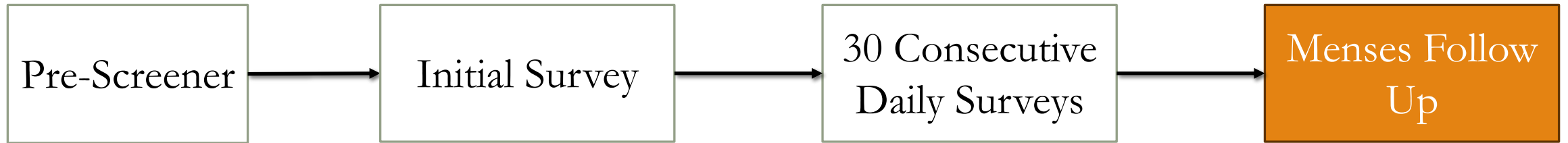
- Women completed an initial ~30-minute online questionnaire
- Women who completed the initial survey were then invited to complete daily surveys

# Study Design



- All items (for current analyses) were identical each day across samples
- All women started (and ended) the daily surveys on identical days
- Completion of  $\geq 10$  daily surveys required for inclusion

# Study Design



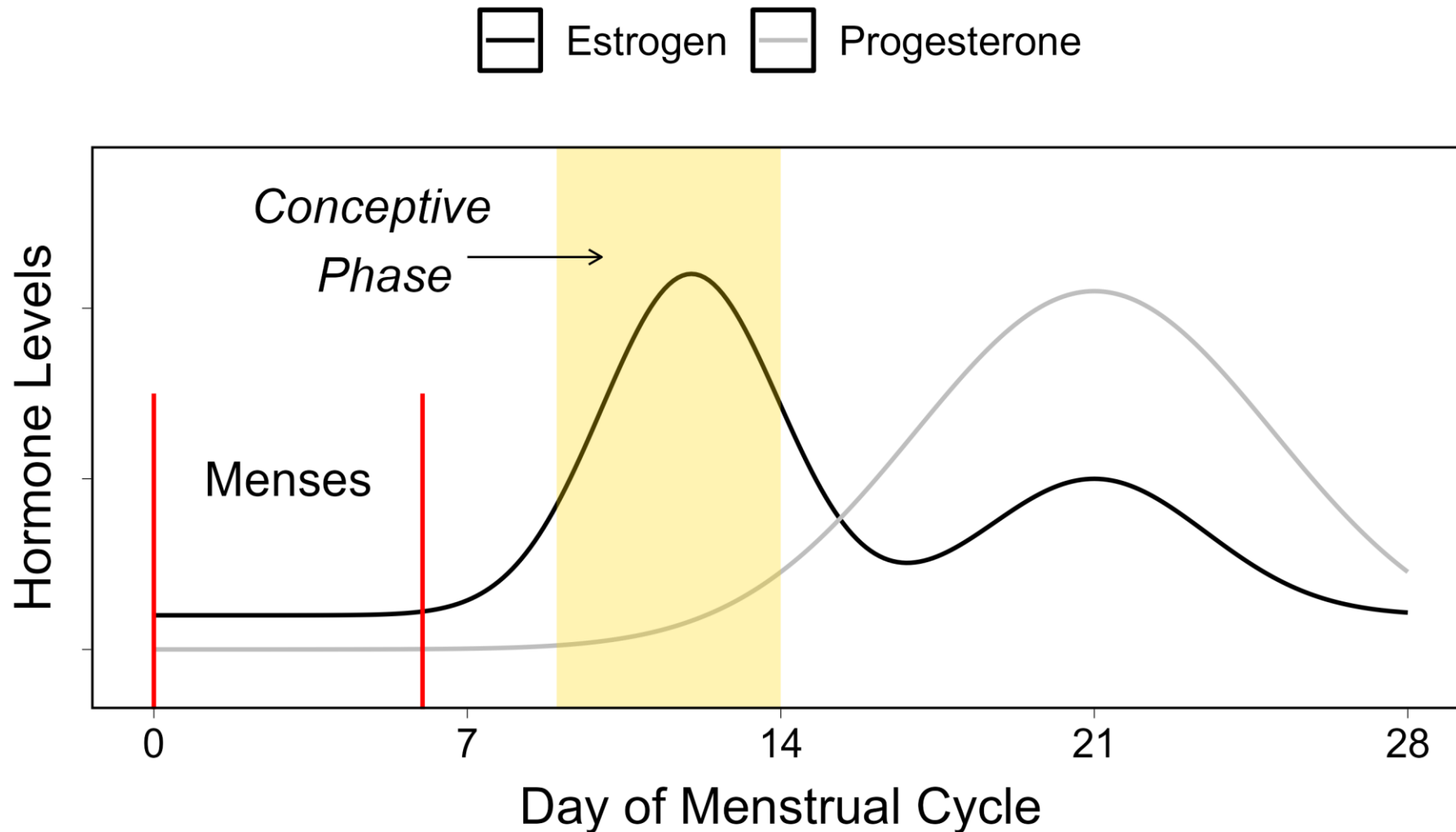
- Follow up survey sent out in waves
- Allowed for calculation of backward count days
- Final Sample: 251/233 women, with a total of 12,475 observations

# Pertinent Measures

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- Fertility
- Primary Partner Sexual Attractiveness
- Extra-pair Sexual Interests (i.e., non-primary partner)

# Tracking Fertility across the Menstrual Cycle: Probability of Conception, Estrogen, and Progesterone



# Estimation of Estradiol, Progesterone, and Probability of Conception

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- Probability of conception: day-specific estimates based on values offered by Arslan et al. (2023), based on Stirnemann et al. (2012).
- Estradiol (E) and Progesterone (P) levels estimated based on day of cycle Arslan et al. (2023).
  - Higher validities of E and P levels from estimation based on cycle day than using salivary assays (based on estimation, E validities  $\sim .6$  to  $.7$  and P validities are  $\sim .7$  to  $.8$ ; Arslan et al., 2023).
- Backward counts were used to estimate 96.4% of 12,475 observations

# Partner and Self Sexual Attractiveness (Initial Survey)

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## Partner Sexual Attractiveness

- “Relative to most men of your partner’s age:”
  - “How sexually attractive is your partner?”
  - “How attractive is your partner?”
  - “How interested are others in your partner as a sexual partner?”
- Response scale: 1 = *much less*, 7 = *much more*
- Reliability and homogeneity for Studies 1/2:  $\alpha = .88/.88$ ,  $\omega_t = .89/.88$ ,  $\omega_c = .98/.99$  (Revelle & Condon, 2025)

**Self Sexual Attractiveness** assessed with same 3 items reworded ( $\alpha = .84/.90$ ,  $\omega_t = .84/.90$ , and  $\omega_c = 1.00/.99$ )

# Daily Extra-Pair Interest Outcome Measure

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- Extra-Pair Attraction (2 items)
  - “I felt strong sexual attraction toward someone other than my current partner”
  - “I was strongly physically attracted to someone other than my current partner.”
- Extra-Pair Sex (1 item scenario)
  - “Imagine that you met a very interesting and attractive person today, and you were sexually attracted to this person, and this person was very sexually attracted to you. Imagine that you have the evening free and you are able to spend the evening with this person. How likely do you think you would be to have sex with this person today, if you could be sure that no one would ever find out about it?”
- Extra-Pair Interest (composite of Z-scored EP Attraction and Z-scored EP Sex)

# As predicted, the attractiveness of partners moderated the effect of hormones

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Probability of Conception X Partner Sexual Attractiveness:

➤  $\gamma = -0.039$  95% *CI* [-0.06, -0.02],  $t(424.6) = -3.74$ ,  $p = .0002$

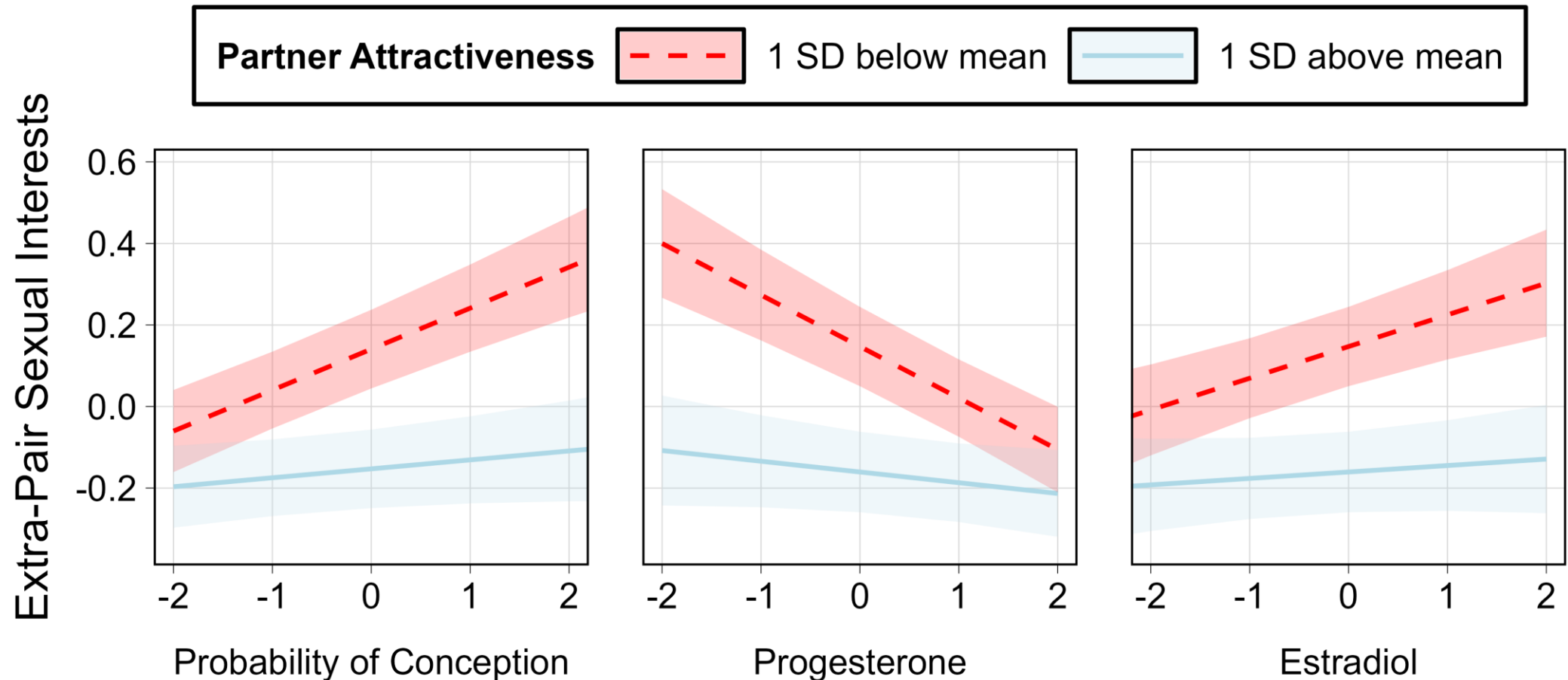
Estradiol X Partner Sexual Attractiveness:

➤  $\gamma = -0.031$  95% *CI* [-0.06, -0.00],  $t(538.7) = -2.31$ ,  $p = .021$

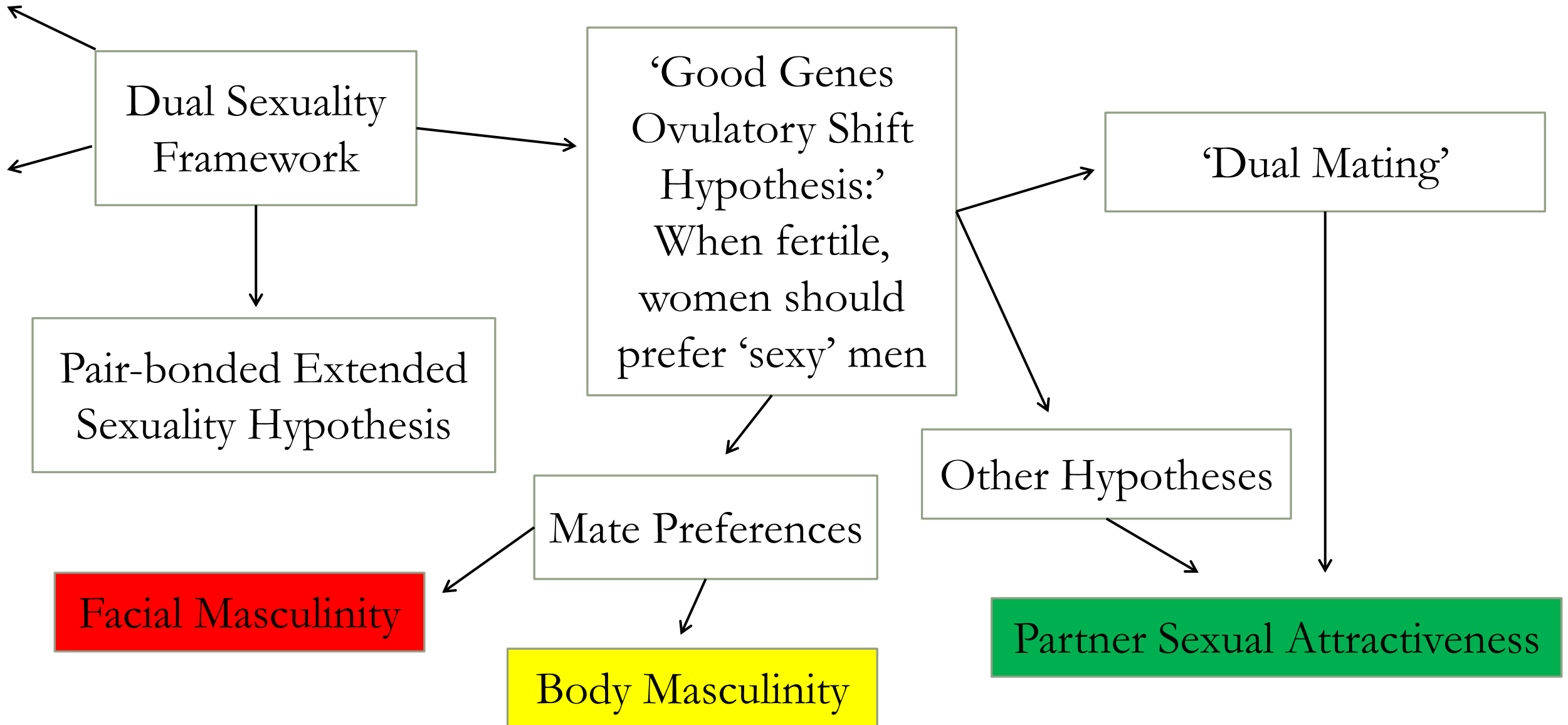
Progesterone X Partner Sexual Attractiveness:

➤  $\gamma = 0.049$  95% *CI* [0.02, 0.08],  $t(429.2) = 3.91$ ,  $p = .0001$

When women had sexually unattractive partners, hormones had an effect on women's extra-pair interests, but when women had attractive partners, hormones had no effect



# What can we conclude?



# In sum:

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- The Partner Attractiveness Moderation effect survived refutation
  - Can future tests (with improved methods) refute it?
  - If not, what explains it?
- Decisive refutations of higher-level theories take time...more work is needed

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Questions?

